Sleep-dependent logical errors

H2-2E emulator via Aqora | Distance-3 X-memory experiment | 2 November 2026

Measured quantity: logical error probability per complete two-round experiment. No division by time has been performed.

What was tested

Code: distance d = 3 rotated surface-code geometry, with nine data qubits and one ancilla reused serially. Each extraction round measures the four X stabilizers. This tests the X-memory sector; Z-stabilizer extraction and arbitrary encoded states were not tested.

Preparation and readout: prepare all nine data qubits in |+> and read them out in the X basis at the end. Record all eight syndrome outcomes and all nine final data outcomes (17 Boolean outputs per shot).

Prepare |+> on nine data qubits
→ X-stabilizer round 1 → all-data sleep block
→ X-stabilizer round 2 → final X readout → offline decoding

Wait: insert one explicit QIR sleep call on each data qubit, with a common raw argument u, between the two extraction rounds. Ten-qubit barriers bound the block; the ancilla has no sleep call. Even u = 0 retains nine explicit sleep calls. Physical units and simultaneous overlap of the nine waits remain unverified.

Sweep: u = 0, 0.1, 0.5, 0.75, 1; 1,000 shots per setting (5,000 total). No artificial Pauli faults or offline measurement flips were added.

Logical error probabilities

Raw argument u	Failures / shots	$P_L(u)$	95% Wilson interval
0	2 / 1000	0.002 (0.20%)	0.05% - 0.73%
0.1	5 / 1000	0.005 (0.50%)	0.21% - 1.17%
0.5	17 / 1000	0.017 (1.70%)	1.06% - 2.71%
0.75	48 / 1000	0.048 (4.80%)	3.64% - 6.31%
1	140 / 1000	0.140 (14.00%)	11.99% - 16.29%

P_L = decoded failures / shots. A failure is the final logical-X parity on data qubits 0, 3, 6 XOR the decoder's predicted logical correction. The probabilities above are not per-round or per-second quantities.

Decoder: fixed minimum-weight perfect matching with assumed p_{data} = 0.002 per round and p_{read} = 0.001, identical across settings. Using p_{data} = 0.02 or 0.1 gave the same failure counts. These assumptions specify decoder weights, not injected noise. All 5,000 packed shots were verified against returned joint counts.

Physical errors and interpretation

Probability notation, raw-argument scaling, and the connection to future timing experiments

Physical data-error probability $p_{data}(u)$

Here $p_{data}(u)$ denotes the **mean final X-readout error probability per data qubit**, averaged across the nine qubits and 1,000 shots. It includes background circuit errors; it is not a directly isolated idle-channel probability. It is distinct from $P_L(u)$.

u = 0: **0.133%**; u = 0.1: **0.400%**; u = 0.5: **3.044%**; u = 0.75: **5.967%**; u = 1: **10.133%**.

$$p_{data}(u) \approx 0.001333 + 0.015245\,u + 0.084495\,u^2$$

u is the raw sleep argument, with no verified physical-time unit. Fitted values are probabilities, not percentages. Valid only as a descriptive interpolation over $0 \le u \le 1$.

This unweighted fit fixes its intercept to the measured u = 0 background. Its coefficients are **not calibrated physical rates**. Unknown pre-existing idle time can turn quadratic dephasing into an apparent linear contribution as well. The fit does not establish an exact noise law, T_2, independent faults, or parallel timing.

Overall error model and what the sweep supports

Jobs requested noisy stabilizer simulation, optimization disabled, and both idle and transport dephasing enabled. Other backend-default errors remained enabled, including gate, preparation, readout and crosstalk faults. Quantinuum documents stochastic Z memory faults with linear-duration and short-time-quadratic contributions in stabilizer mode [1].

The inter-round mean detector probability rises from **0.450% to 23.975%**, while the first-round mean stays between 0.250% and 0.500%. This localizes the dominant increase across the inserted wait. The physical-error curvature is consistent with a substantial quadratic contribution. The u = 0.1 logical comparison is inconclusive at this shot count; the larger waits show clear increases.

The data do not uniquely isolate idle dephasing from transport, background faults, scheduling changes and decoder effects. Confirmed units and timing plus matched noise-channel controls are needed to infer physical idle rates. One job per setting does not measure run-to-run variation.

Probability now; time-normalized metric later

For a future R-round experiment with a physical wait Δt between adjacent rounds, measure $P_L(R, \Delta t)$ and the total duration T. The planned time-normalized metric is $\mathbf{r_L^{eff} = P_L(R, \Delta t) / T}$, with units of inverse time. Specify whether T includes preparation/readout and extraction time; storage-and-extraction time is approximately $R\tau_{syn} + (R - 1)\Delta t$. This metric is not automatically a constant-hazard parameter, and the present two-round probabilities cannot simply be extrapolated to many rounds.

[1] Quantinuum: Emulator Noise Model and H2 Emulator Defaults.
Evidence: saved Aqora H2-2E five-point run, build 67d11d0348065703; decoded-count and noise-characterization JSON records in SG_final_round/results/.

Conclusion

Increasing the raw sleep argument produces a pronounced increase in physical data-error probability and decoded logical error probability in this d = 3, two-round X-memory experiment. The growth is compatible with the emulator's memory-noise model, but does not establish a pure idle-noise rate, physical sleep units, or simultaneous waits. Future many-round experiments will report time-normalized logical error only after total physical duration is defined and verified.